

Chapter

Periodic Properties And Variations Of Properties

1. ${}^1_1\text{H}^2$ is an isotope of hydrogen. In the modern periodic table it will: SQP 2025

- (a) be placed before hydrogen
- (b) be placed after hydrogen
- (c) be placed at the same position as hydrogen
- (d) not have any position in the periodic table.

2. In the 2nd period, Neon has maximum ionisation potential because:

- (a) It has unstable electronic configuration. **MAIN 2024**
- (b) It easily accepts electrons.
- (c) It easily loses electrons.
- (d) The outer most shell is completely filled.

3. Electron Affinity is maximum in: MAIN 2024

- (a) Mg
- (b) Ar
- (c) Li
- (d) Br

4. In a periodic table, the element of period 3 are arranged in the increasing order of ionisation potential as SEM-I 2021

- (a) B, N, Cl, Ar
- (b) Mg, Si, S, Ar
- (c) Ar, Si, S, Mg
- (d) Si, Ar, Cl, Mg

5. The tendency of an atom to attract shared pair of electrons to itself when forming a chemical bond is known as SEM-I 2021

- (a) electron affinity
- (b) electronegativity
- (c) ionisation potential
- (d) nuclear charge

6. Alkaline earth metals have the same. SEM-I 2021

- (a) number of valence electrons
- (b) number of shells
- (c) metallic property
- (d) ionisation potential

7. Element with an atomic number 19 will. SEM-I 2021

- (a) accept an electron and get oxidised
- (b) accept an electron and get reduced
- (c) lose an electron and get oxidised
- (d) lose an electron and get reduced

8. If an element has low ionisation potential, then it is likely to be a. SEM-I 2021

- (a) metal
- (b) metalloid
- (c) non-metal
- (d) inert gas

9. The non-metallic properties of elements from left to right in a periodic table

- (a) increases SEM-I 2021
- (b) decreases
- (c) remains same
- (d) first increases and then decreases

10. An element in period 2 whose electron affinity is zero: SQP 2022

- (a) Carbon
- (b) Neon

- (c) Sodium
- (d) Sulphur

11. The most active metals are present in 1 and 2 of the Periodic Table. COMP 2021

- (a) Period
- (b) Group
- (c) Both (a) and (b)
- (d) None of these

12. The correct order of electron affinities of N, O, S and Cl is MAIN 2023

- (a) $O < S < Cl < N$
- (b) $O < N < Cl < S$
- (c) $O = Cl < N = S$
- (d) $N < O < S < Cl$

13. The tendency of an atom to attract shared pair of electrons to itself when combined in a compound is called along the period. MAIN 2023

- (a) electro-positivity
- (b) electro-negativity
- (c) inert
- (d) none of the above

14. The non-metallic character of the elements down the group. SQP 2023

- (a) remains the same
- (b) increases
- (c) decreases
- (d) depends on the period

15. Which of the following statements is not a correct statement about the trends when going from left to right across the Periodic Table? MAIN 2021

- (a) The elements become less metallic in nature.
- (b) The atoms lose their electrons more easily.

- (c) The number of valency electrons increases.
- (d) The oxides become more acidic.

16. Elements in a period, all have the same number of in their atoms.

- (a) electron shells **MAIN 2023**
- (b) valence electrons
- (c) valency
- (d) None of the above

17. Down the group, electron affinity SQP 2021

- (a) remains same
- (b) decreases
- (c) increases
- (d) none of the above

18. The amount of energy released when one or more electron is added to the neutral atom. COMP 2023

- (a) Ionisation energy
- (b) Electron affinity
- (c) Electro-negativity
- (d) Atomicity

19. Lithium exhibits diagonal relationship with. SQP 2021

- (a) Calcium
- (b) Barium
- (c) Magnesium
- (d) Beryllium

20. Horizontal rows of periodic table are called. MAIN 2022

- (a) Groups
- (b) Compounds
- (c) Elements
- (d) Periods

21. The non-metals are present COMP 2021

- (a) On the left hand side of the periodic table.
- (b) In the middle of the periodic table.
- (c) On the right hand side of the periodic table.
- (d) In the centre of the periodic table

22. On moving from left to right in a given period, the number of shells

..... COMP 2023

- (a) increases
- (b) decreases
- (c) remains the same
- (d) none of the above

23. An element with the atomic number 19 will most likely combine chemically with the element whose atomic number is : SQP 2004

- (a) 18
- (b) 11
- (c) 17
- (d) 20

24. Down the group, electron affinity ? COMP 2021

- (a) decreases
- (b) increases
- (c) remains same
- (d) None of these

25. The alkali metal in the period 2. SQP 2021

- (a) Beryllium
- (b) Boron
- (c) Lithium
- (d) Carbon

26. Ionisation Potential increases over a period from left to right because the :

- (a) Atomic radius increases and nuclear charge increase **COMP 2023**
- (b) Atomic radius decreases and nuclear charge increase
- (c) Atomic radius increases and nuclear charge decreases
- (d) Atomic radius and nuclear charge decrease

27. The elements having the least number of electrons shells: MAIN 2021

- (a) F
- (b) Li
- (c) Na
- (d) K

28. Chlorine has electron affinity than fluorine. SQP 2021

- (a) less
- (b) same
- (c) more
- (d) none of the above

29. Fifth period contains elements. COMP 2023

- (a) 6
- (b) 18
- (c) 12
- (d) 32

30. Assertion : Chlorine is called halogen. **MAIN 2021**

Reason : It is because chlorine reacts with metals to form salts.

- (a) Assertion and reason both are correct statements and reason is the correct explanation of the assertion.
- (b) Assertion and reason both are correct statements, but reason is not the correct explanation of the assertion.
- (c) Assertion is false, but reason is true
- (d) Assertion is true, but reason is false

31. A non-metal which has three electrons in the valence shell. SQP 2023

32. The most electronegative element in the third period. MAIN 2021

33. Name an inert gas having two electrons in its outermost orbit. COMP 2023

34. If an element has one electron in the outermost shell, then it is likely _____ to have the [smallest/largest] atomic size amongst all the elements in the same period. SQP 2025

35. Elements of Group 1 are called _____. MAIN 2020

36. There are _____ groups and _____ periods in the periodic table. SQP 2023

37. Metals form _____ chlorides COMP 2021

38. The electron affinity of an element X is greater than that of element Y.

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(i) How is the oxidising power of X likely to compare with that of Y?

(ii) How is the electronegativity of X likely to compare with that of Y?

(iii) State whether X is likely to be placed to the left or to the right of Y in the periodic table?

39. Identify the following: MAIN 2024

(a) An element in period 1 which can be placed in both group 1 and group 17 of the periodic Table.

(b) The element having electronic configuration 2, 8, 6.

(c) The most electronegative element of period 3.

40. Explain clearly, why atomic number of an element is most important to the chemist than its relative atomic mass. MAIN 2023

41. Explain why reducing power of elements increases as one goes down a group ? SQP 2022

Solution

1. (c) — *be placed at the same position as hydrogen.*

${}^{12}\text{H}$ is an isotope of hydrogen.

In the modern periodic table, **all isotopes of an element** are placed in the same **position** because their atomic number is the same.

So, ${}^{12}\text{H}$ stays with hydrogen.

2. (d) — *The outermost shell is completely filled.*

Neon has a **stable electronic configuration** with a **completely filled outermost shell** (octet). Its electrons are held tightly, so it has the **maximum ionisation potential** in the 2nd period.

3. (d) — *Br (Bromine)*

Electron affinity increases across a period and decreases down a group.

Bromine (Br), being a halogen, has a strong tendency to **gain an electron** to complete its octet, giving it the **maximum electron affinity** among the given elements.

4. Thus (b) is correct option.

5. (b) — *Electronegativity*

Electronegativity is the tendency of an atom to **attract the shared pair of electrons** towards itself in a chemical bond.

6. (a) — *Number of valence electrons*

Alkaline earth metals have **2 valence electrons** in their outermost s-orbital, giving them similar chemical properties.

7. (c) — *Lose an electron and get oxidised*

The element with atomic number 19 is **potassium (K)**, a Group I element. It **loses one electron** from its outermost shell to form a cation and **gets oxidised**.

8. (a) — *Metal*

An element with **low ionisation potential** easily **loses electrons** to form cations, which is a characteristic property of **metals**.

9. (a) — *Increases*

From **left to right** in a period, **atomic size decreases** and **nuclear charge increases**, making it easier to gain electrons, so **non-metallic character increases**.

10. An element in period 2 whose electron affinity is zero: **(b) Neon**

Neon is a **noble gas** with a **completely filled valence shell**. It does not easily accept electrons, so its **electron affinity is nearly zero**.

11. The most active metals are present in: **(b) Group 1 and 2**

Highly reactive metals belong to **Groups I A and II A** of the periodic table and easily react with oxygen and water.

12. The correct order of electron affinities of N, O, S, and Cl is:

(d) $N < O < S < Cl$

- Electron affinity **increases across a period** and **decreases down a group**.
- **Sulphur** has higher electron affinity than **oxygen** due to its larger size and lower electron-electron repulsion.
- Therefore, the order is: **$N < O < S < Cl$** .

13. (b) Electronegativity

Electronegativity is the ability of an atom to **attract bonding electrons** towards itself in a compound.

14. (c) Decreases

Down a group, **atomic size increases** and atoms **lose electrons more easily**, so **non-metallic character decreases**.

15. (b) The atoms lose their electrons more easily

Across a period, **nuclear charge increases** while the number of shells remains the same, making it **harder for atoms to lose electrons**.

16. (a) Electron shells

All elements in a **period** have the **same number of electron shells**, though their **valence electrons** increase across the period.

17. (b) Decreases

As we move **down a group**, atoms become larger, so the added electron is **farther from the nucleus**, resulting in **lower electron affinity**.

18. (b) Electron affinity

Electron affinity is the energy released when an **electron is added** to a neutral atom. Higher energy released means higher electron affinity.

19. (c) Magnesium

Lithium and **Magnesium** show similar **chemical properties** due to comparable **charge density** and **small atomic size**, demonstrating a **diagonal relationship**.

20. (d) Periods

In the periodic table, **horizontal rows** are called **periods**, and **vertical columns** are called **groups**.

21. (c) On the right hand side of the periodic table

Non-metals are strong **oxidizing agents** and are located on the **right side** of the periodic table.

22. (c) Remains the same

Across a period, **valence electrons increase**, but the **number of electron shells stays the same**.

23. (c) 17

- Element 19 (**potassium, K**) has **1 valence electron**.
- Element 17 (**chlorine, Cl**) needs **1 electron to complete its octet**.
- Hence, **K and Cl combine to form KCl**.

24. (a) Decreases

As we move **down a group**, atoms become larger, so the added electron is **farther from the nucleus**, resulting in **lower electron affinity**.

25. (c) Lithium

Period 2 elements: Li, Be, B, C, N, O, F, Ne.

Lithium (Li) is the **alkali metal** in this period.

26. (b) Atomic radius decreases and nuclear charge increases

Across a period, **atomic radius decreases** and **nuclear charge increases**, so electrons are held more tightly, resulting in **higher ionisation energy**.

27. (b) Li

- **Li (Lithium)** is in **period 2**, so it has **2 electron shells**.
- **F (period 2)** also has 2 shells, but Li has **fewer electrons overall**, and among the options, Li represents the **alkali metal with the simplest structure**.

28. (c) More

- Fluorine is smaller, so added electrons experience **greater repulsion** in its compact shell.
- Chlorine has a **larger atomic size**, so it can **accept an electron more easily**, giving it **higher electron affinity**.

29. Thus (b) is correct option.

30. (c) — *Assertion is false, but reason is true*

- The **reason** is true: chlorine reacts with metals to form salts.
- However, the **assertion** as phrased may be considered false if interpreted incorrectly, but generally, chlorine **is a halogen**.

31. Boron

32. Chlorine

33. Helium

34. Largest

35. Alkali metals

36. Eighteen, seven

37. Ionic

38. (i) **Oxidising power:** X has **more oxidising power** than Y.

(ii) **Electronegativity:** X is **more electronegative** than Y.

(iii) **Position in periodic table:** X is placed **to the right** of Y.

39. (a) Hydrogen (H) (b) Sulphur (S) (c) Chlorine (Cl)

40. This is because atomic number being equal to the number of electrons in an atom of the element, it enables him to write down the electronic configurations of elements. The elements having similar electronic configurations can then be put together in the same group. This helps in the systematic classification of elements. On the other hand, the atomic mass of an element does not tell us anything about the number of electrons present in one atom of the element.

41. When an atom loses an electron/electrons from its valence shell, the element is said to be reducing agent and the property of the element is called reducing property. The reducing power of an element depends upon the ease with which it can lose electrons. In case of elements at the bottom of a group, the valence electrons are held very loosely by the nucleus. Thus, such

elements easily lose their valence electrons and hence have a higher reducing power.